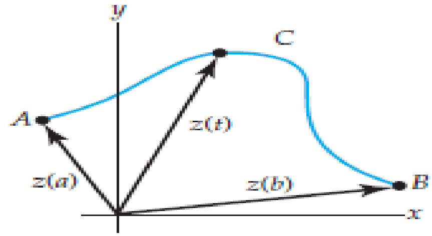
Complex Integrals



$z(t) = x(t) + iy(t)$ as a position vector

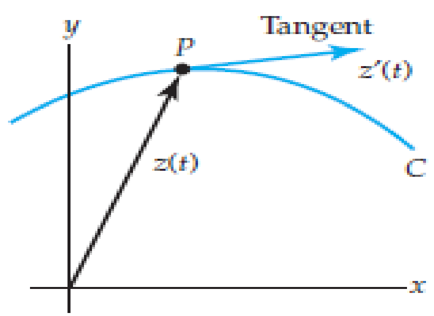
Suppose the continuous real-valued functions $x = x(t)$, $y = y(t)$, $a \le t \le b$, are parametric equations of a curve C in the complex plane. If we use these equations as the real and imaginary parts in $z = x + iy$, we saw that we can describe the points z on C by means of a complex-valued function of a real variable t called a **parametrization of C**:

$$z(t) = x(t) + iy(t), \ a \le t \le b. \tag{1}$$

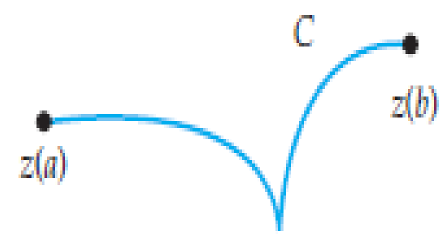
For example, the parametric equations $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$, describe a unit circle centered at the origin. A parametrization of this circle is $z(t) = \cos t + i \sin t$, or $z(t) = e^{it}, 0 \le t \le 2\pi$.

The point $z(a) = x(a) + iy(a)$ or $A = (x(a), \ y(a))$ is called the **initial point** of C and $z(b) = x(b) + iy(b)$ or $B = (x(b), y(b))$ is its **terminal point**. We also saw that $z(t) = x(t) + iy(t)$ could also be interpreted as a two-dimensional vector function. Consequently, $z(a)$ and $z(b)$ can be interpreted as position vectors.

Contours The notions of curves in the complex plane that are smooth, piecewise smooth, simple, closed, and simple closed are easily formulated in terms of the vector function (1). Suppose the derivative of (1) is $z'(t) = x'(t) + iy'(t)$. We say a curve C in the complex plane is **smooth** if $z'(t)$ is continuous and never zero in the interval $a \leq t \leq b$. As shown Figure , since the vector $z'(t)$ is not zero at any point P on C, the vector $z'(t)$ is tangent to C at P. In other words, a smooth curve has a continuously turning tangent; put yet another way, a smooth curve can have no sharp corners or cusps. **A piecewise smooth curve** C has a continuously turning tangent, except possibly at the points where the component smooth curves $C_1, C_2, \ldots, C_n$ are joined together. A curve C in the complex plane is said to be a **simple** if $z(t_1) \neq z(t_2)$ for $t_1 \neq t_2$, except possibly for $t = a$ and $t = b$. C is a **closed curve** if $z(a) = z(b)$. C is a **simple closed curve** if $z(t_1) \neq z(t_2)$ for $t_1 \neq t_2$ and $z(a) = z(b)$. In complex analysis, a piecewise smooth curve C is called a **contour** or **path**.



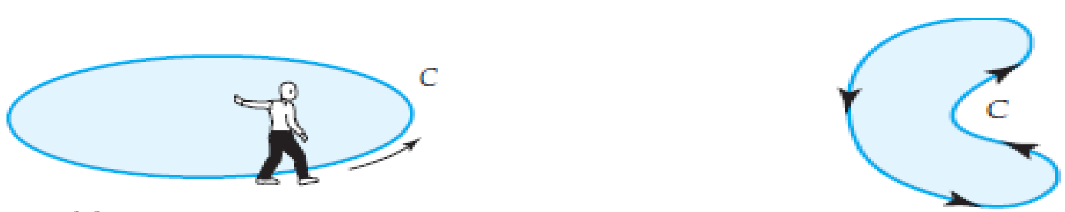
$z'(t) = x'(t) + iy'(t)$ as a tangent vector



Curve C is not smooth since it has a cusp.

Just as we did in the preceding section, we define the **positive direction** on a contour C to be the direction on the curve corresponding to increasing values of the parameter t. It is also said that the curve C has **positive orientation.** In the case of a simple closed curve C, the positive direction roughly corresponds to the counterclockwise direction or the direction that a person must walk on C in order to keep the interior of C to the left. For example, the circle $z(t) = e^{it}$, $0 \leq t \leq 2\pi$, has positive orientation. See Figure

The **negative direction** on a contour C is the direction opposite the positive direction. If C has an orientation, the **opposite curve**, that is, a curve with opposite orientation, is denoted by $-C$. On a simple closed curve, the negative direction corresponds to the clockwise direction.



(a) Positive direction

(b) Positive direction

Interior of each curve is to

the left.

Complex Integral An integral of a function f of a complex variable z that is defined on a contour C is denoted by $\int_C f(z)\,dz$ and is called a **complex** integral

Steps Leading to the Definition of the Complex Integral

1. *Let f be a function of a complex variable z defined at all points on a smooth curve C that lies in some region of the complex plane. Let C be defined by the parametrization $z(t) = x(t) + iy(t)$, $a \le t \le b$.*

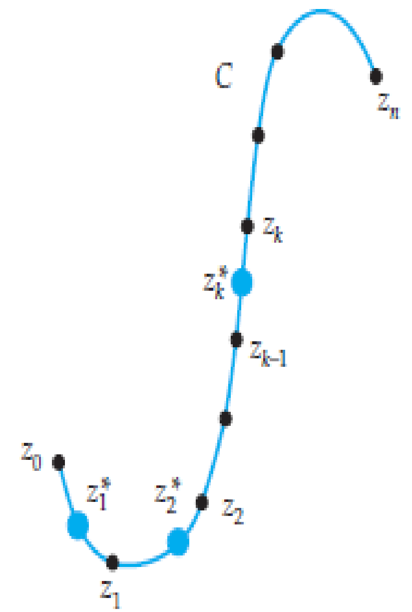
2. *Let P be a partition of the parameter interval $[a, b]$ into n subintervals $[t_{k-1}, t_k]$ of length $\Delta t_k = t_k - t_{k-1}$:*

$$a = t_0 < t_1 < t_2 < \cdots < t_{n-1} < t_n = b.$$

The partition P induces a partition of the curve C into n subarcs whose initial and terminal points are the pairs of numbers

$$\begin{aligned} z_0 &= x(t_0) + iy(t_0), & z_1 &= x(t_1) + iy(t_1), \\ z_1 &= x(t_1) + iy(t_1), & z_2 &= x(t_2) + iy(t_2), \\ &\vdots & &\vdots \\ z_{n-1} &= x(t_{n-1}) + iy(t_{n-1}), & z_n &= x(t_n) + iy(t_n). \end{aligned}$$

Let $\Delta z_k = z_k - z_{k-1}$, $k = 1, 2, \ldots, n$. See Figure



Partition of curve C into n subarcs is induced by a partition P of the parameter interval $[a, b]$.

3. *Let $\|P\|$ be the norm of the partition P of $[a, b]$, that is, the length of the longest subinterval.*

4. *Choose a point $z_k^* = x_k^* + iy_k^*$ on each subarc of C.* .

5. *Form n products $f(z_k^*)\Delta z_k$, $k = 1, 2, \ldots, n$, and then sum these products:*

$$\sum_{k=1}^{n} f(z_k^*)\Delta z_k.$$

Definition [redacted] Complex Integral

The complex integral of f on C is

$$\int_C f(z)\,dz = \lim_{\|P\|\to 0} \sum_{k=1}^{n} f(z_k^*)\,\Delta z_k. \tag{2}$$

If the limit in (2) exists, then f is said to be **integrable on** C. The limit exists whenever if f is continuous at all points on C and C is either smooth or piecewise smooth. Consequently we shall, hereafter, assume these conditions as a matter of course. Moreover, we will use the notation $\oint_C f(z)\,dz$ to represent a complex integral around a *positively oriented* closed curve C. When it is important to distinguish the direction of integration around a closed curve, we will employ the notations

$$\ointctrclockwise_C f(z)\,dz \quad \text{and} \quad \ointclockwise_C f(z)\,dz$$

to denote integration in the positive and negative directions, respectively.

Complex-Valued Function of a Real Variable Before turning to the properties of contour integrals and the all-important question of how to evaluate a contour integral, we need to digress briefly to enlarge upon the concept of a **complex-valued function of a real variable** introduced As already mentioned, a parametrization of a curve C of the form given in (1) is a case in point. Let's consider another simple example. If t represents a real variable, then the output of the function $f(t) = (2t+i)^2$ is a complex number. For $t = 2$,

$$f(2) = (4+i)^2 = 16 + 8i + i^2 = 15 + 8i.$$

In general, if f_1 and f_2 are real-valued functions of a real variable t (that is, real functions), then $f(t) = f_1(t) + if_2(t)$ is a complex-valued function of a real variable t. What we are really interested in at the moment is the definite integral $\int_a^b f(t)\,dt$, in other words, integration of complex-valued function $f(t) = f_1(t) + if_2(t)$ of real variable t carried out over a real interval. Continuing with the specific function $f(t) = (2t+i)^2$ it seems logical to write on, say, the interval $0 \le t \le 1$,

$$\int_0^1 (2t+i)^2 dt = \int_0^1 (4t^2 - 1 + 4ti)\,dt = \int_0^1 (4t^2-1)dt + i\int_0^1 4t\,dt. \qquad (3)$$

The integrals $\int_0^1 (4t^2-1)dt$ and $\int_0^1 4t\,dt$ in (3) are real, and so one would be inclined to call them the real and imaginary parts of $\int_0^1 (2t+i)^2 dt$. Each of these real integrals can be evaluated using the fundamental theorem of calculus

$$\int_0^1 (4t^2-1)dt = \left(\frac{4}{3}t^3 - t\right)\Bigg|_0^1 = \frac{1}{3} \quad \text{and} \quad \int_0^1 4t\,dt = 2t^2\Big|_0^1 = 2.$$

Thus (3) becomes $\int_0^1 (2t+i)^2 dt = \frac{1}{3} + 2i$.

If f_1 and f_2 are real-valued functions of a real variable t continuous on a common interval $a \le t \le b$, then we *define* the integral of the complex-valued function $f(t) = f_1(t) + i f_2(t)$ on $a \le t \le b$ in terms of the definite integrals of the real and imaginary parts of f:

$$\int_a^b f(t)\,dt = \int_a^b f_1(t)\,dt + i\int_a^b f_2(t)\,dt. \tag{4}$$

The continuity of f_1 and f_2 on $[a,\ b]$ guarantees that both $\int_a^b f_1(t)\,dt$ and $\int_a^b f_2(t)\,dt$ exist.

All of the following familiar properties of integrals can be proved directly from the definition given in (4). If $f(t) = f_1(t)+if_2(t)$ and $g(t) = g_1(t)+ig_2(t)$ are complex-valued functions of a real variable t continuous on an interval $a \le t \le b$, then

$$\int_a^b k\, f(t)\, dt = k \int_a^b f(t)\, dt, \quad k \text{ a complex constant,} \tag{5}$$

$$\int_a^b (f(t) + g(t))\, dt = \int_a^b f(t)\, dt + \int_a^b g(t)\, dt, \tag{6}$$

$$\int_a^b f(t)\, dt = \int_a^c f(t)\, dt + \int_c^b f(t)\, dt, \tag{7}$$

$$\int_b^a f(t)\, dt = -\int_a^b f(t)\, dt. \tag{8}$$

Evaluation of Contour Integrals To facilitate the discussion on how to evaluate a contour integral $\int_C f(z)\,dz$, let us write (2) in an abbreviated form. If we use $u+iv$ for f, $\Delta x+i\Delta y$ for Δz, lim for $\lim\limits_{||P||\to 0}$, $\sum$ for $\sum_{k=1}^{n}$ and then suppress all subscripts, (2) becomes

$$\int_C f(z)dz = \lim \sum (u+iv)(\Delta x + i\Delta y)$$
$$= \lim \left[\sum (u\Delta x - v\Delta y) + i\sum (v\Delta x + u\Delta y)\right].$$

The interpretation of the last line is

$$\int_C f(z)\,dz = \int_C u\,dx - v\,dy + i\int_C v\,dx + u\,dy. \tag{9}$$

In other words, the real and imaginary parts of a contour integral $\int_C f(z)\,dz$ are a pair of real line integrals $\int_C u\,dx - v\,dy$ and $\int_C v\,dx + u\,dy$. Now if $x = x(t)$, $y = y(t)$, $a \le t \le b$ are parametric equations of C, then $dx = x'(t)\,dt$, $dy = y'(t)\,dt$. By replacing the symbols x, y, dx, and dy by $x(t)$, $y(t)$, $x'(t)\,dt$, and $y'(t)\,dt$, respectively, the right side of (9) becomes

$$
\begin{aligned}
&\overbrace{\int_a^b [u(x(t),y(t))\; x'(t) - v(x(t),y(t))\; y'(t)]\,dt}^{\int_C u dx - v dy} \\
&\quad + \; i \overbrace{\int_a^b [v(x(t),y(t))\; x'(t) + u(x(t),y(t))\; y'(t)]\,dt.}^{\int_C v dx + u dy}
\end{aligned} \tag{10}
$$

If we use the complex-valued function (1) to describe the contour C, then (10) is the same as $\int_a^b f(z(t))\, z'(t)\, dt$ when the integrand

$$
f(z(t))\, z'(t) = [u(x(t),y(t)) + iv(x(t),y(t))]\, [x'(t) + iy'(t)]
$$

is multiplied out and $\int_a^b f\, (z(t))\;\, z'(t)\, dt$ is expressed in terms of its real and imaginary parts. Thus we arrive at a practical means of evaluating a contour integral.

Theorem Evaluation of a Contour Integral

If f is continuous on a smooth curve C given by the parametrization $z(t) = x(t) + iy(t)$, $a \le t \le b$, then

$$\int_C f(z)\,dz = \int_a^b f(z(t))\,z'(t)\,dt. \tag{11}$$

EXAMPLE 1 Evaluating a Contour Integral

Evaluate $\int_C \bar{z}\,dz$, where C is given by $x = 3t$, $y = t^2$, $-1 \le t \le 4$.

Solution From (1) a parametrization of the contour C is $z(t) = 3t + it^2$. Therefore, with the identification $f(z) = \bar{z}$ we have $f(z(t)) = \overline{3t + it^2} = 3t - it^2$. Also, $z'(t) = 3 + 2it$, and so by (11) the integral is

$$\int_C \bar{z}\,dz = \int_{-1}^{4} (3t - it^2)(3 + 2it)\,dt = \int_{-1}^{4} \left[2t^3 + 9t + 3t^2 i\right] dt.$$

Now in view of (4), the last integral is the same as

$$\begin{aligned}\int_C \bar{z}\,dz &= \int_{-1}^{4} (2t^3 + 9t)\,dt + i\int_{-1}^{4} 3t^2\,dt \\ &= \left(\frac{1}{2}t^4 + \frac{9}{2}t^2\right)\Big|_{-1}^{4} + it^3\Big|_{-1}^{4} = 195 + 65i.\end{aligned}$$

EXAMPLE 2 Evaluating a Contour Integral

Evaluate $\oint_C \frac{1}{z}\,dz$, where C is the circle $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$.

Solution In this case $z(t) = \cos t + i\sin t = e^{it}$, $z'(t) = ie^{it}$, and $f(z(t)) = \frac{1}{z(t)} = e^{-it}$. Hence,

$$\oint_C \frac{1}{z}\,dz = \int_0^{2\pi} \left(e^{-it}\right) ie^{it}\,dt = i\int_0^{2\pi} dt = 2\pi i.$$

Theorem **Properties of Contour Integrals**

Suppose the functions f and g are continuous in a domain D, and C is a smooth curve lying entirely in D. Then

(*i*) $\int_C kf(z)\,dz = k\int_C f(z)\,dz$, k a complex constant.

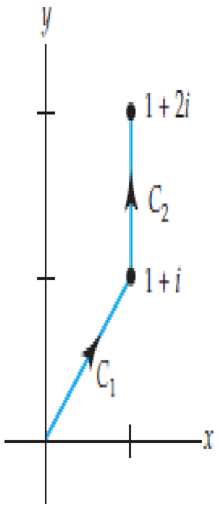
(*ii*) $\int_C [f(z) + g(z)]\,dz = \int_C f(z)\,dz + \int_C g(z)\,dz$.

(*iii*) $\int_C f(z)\,dz = \int_{C_1} f(z)\,dz + \int_{C_2} f(z)\,dz$, where C consists of the smooth curves C_1 and C_2 joined end to end.

(*iv*) $\int_{-C} f(z)\,dz = -\int_C f(z)\,dz$, where $-C$ denotes the curve having the opposite orientation of C.

EXAMPLE 3 C Is a Piecewise Smooth Curve

Evaluate $\int_C (x^2 + iy^2)\,dz$, where C is the contour shown in Figure



Contour C is piecewise smooth.

Solution In view of Theorem (*iii*) we write

$$\int_C (x^2 + iy^2)\,dz = \int_{C_1} (x^2 + iy^2)\,dz + \int_{C_2} (x^2 + iy^2)\,dz.$$

Since the curve C_1 is defined by $y = x$, it makes sense to use x as a parameter. Therefore, $z(x) = x + ix$, $z'(x) = 1 + i$, $f(z) = x^2 + iy^2$, $f(z(x)) = x^2 + ix^2$, and

$$\begin{aligned}\int_{C_1} (x^2 + iy^2)\,dz &= \int_0^1 \overbrace{(x^2 + ix^2)}^{(1+i)x^2}(1 + i)\,dx \\ &= (1+i)^2 \int_0^1 x^2 dx = \frac{(1+i)^2}{3} = \frac{2}{3}i. \qquad (12)\end{aligned}$$

The curve C_2 is defined by $x = 1$, $1 \le y \le 2$. If we use y as a parameter, then $z(y) = 1 + iy$, $z'(y) = i$, $f(z(y)) = 1 + iy^2$, and

$$\int_{C_2} (x^2 + iy^2)\,dz = \int_1^2 (1 + iy^2)i\,dy = -\int_1^2 y^2\,dy + i\int_1^2 dy = -\frac{7}{3} + i. \quad (13)$$

Combining (10) and (13) gives $\int_C (x^2 + iy^2)\,dz = \frac{2}{3}i + (-\frac{7}{3} + i) = -\frac{7}{3} + \frac{5}{3}i.$

The length of a plane curve

curve is $L = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt$. But if $z'(t) = x'(t) + iy'(t)$, then $|z'(t)| = \sqrt{[x'(t)]^2 + [y'(t)]^2}$ and, consequently, $L = \int_a^b |z'(t)|\, dt$.

Theorem A Bounding Theorem

If f is continuous on a smooth curve C and if $|f(z)| \leq M$ for all z on C, then $\left|\int_C f(z)\, dz\right| \leq ML$, where L is the length of C.

EXAMPLE 4 **A Bound for a Contour Integral**

Find an upper bound for the absolute value of $\oint_C \frac{e^z}{z+1}\,dz$ where C is the circle $|z| = 4$.

Solution First, the length L (circumference) of the circle of radius 4 is 8π.
for all points z on the circle that $|z+1| \geq |z| - 1 = 4 - 1 = 3$. Thus

$$\left|\frac{e^z}{z+1}\right| \leq \frac{|e^z|}{|z|-1} = \frac{|e^z|}{3}. \tag{15}$$

In addition, $|e^z| = |e^x(\cos y + i\sin y)| = e^x$. For points on the circle $|z| = 4$, the maximum that $x = \mathrm{Re}(z)$ can be is 4, and so (15) yields

$$\left|\frac{e^z}{z+1}\right| \leq \frac{e^4}{3}.$$

From the ML-inequality (Theorem) we have

$$\left|\oint_C \frac{e^z}{z+1}\,dz\right| \leq \frac{8\pi e^4}{3}.$$

There is no unique parametrization for a contour C. You should verify that

$$z(t) = e^{it} = \cos t + i \sin t,\ 0 \le t \le 2\pi$$
$$z(t) = e^{2\pi it} = \cos 2\pi t + i \sin 2\pi t,\ 0 \le t \le 1$$
$$z(t) = e^{\pi it/2} = \cos \frac{\pi t}{2} + i \sin \frac{\pi t}{2},\ 0 \le t \le 4$$

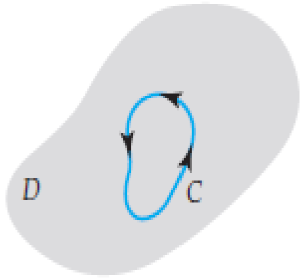
are all parametrizations, oriented in the positive direction, for the unit circle $|z| = 1$.

Cauchy-Goursat Theorem

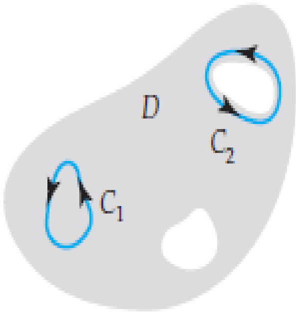
Simply and Multiply Connected Domains

Recall that a domain is an open connected set in the complex plane. We say that a domain D is **simply connected** if every simple closed contour C lying entirely in D can be shrunk to a point without leaving D.

In other words, if we draw any simple closed contour C so that it lies entirely within a simply connected domain, then C encloses only points of the domain D. Expressed yet another way, a simply connected domain has no "holes" in it. The entire complex plane is an example of a simply connected domain; the annulus defined by $1 < |z| < 2$ is not simply connected. (Why?) A domain that is not simply connected is called a **multiply connected domain**; that is, a multiply connected domain has "holes" in it. Note in Figure that if the curve C_2 enclosing the "hole" were shrunk to a point, the curve would have to leave D eventually. We call a domain with one "hole" **doubly connected**, a domain with two "holes" **triply connected**, and so on. The open disk defined by $|z| < 2$ is a simply connected domain; the open circular annulus defined by $1 < |z| < 2$ is a doubly connected domain.



Simply connected domain D



Multiply connected domain D

Cauchy's Theorem

Suppose that a function f is analytic in a simply connected domain D and that f' is continuous in D. Then for every simple closed contour C in D, $\oint_C f(z)\,dz = 0$. (1)

If the real-valued functions $P(x, y)$ and $Q(x, y)$ along with their first-order partial derivatives are continuous on a domain that contains C and R, then

$$\oint_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA. \tag{2}$$

$$\begin{aligned}\oint_C f(z)\,dz &= \oint_C u(x,y)\,dx - v(x,y)\,dy + i\oint_C v(x,y)\,dx + u(x,y)\,dy \\ &= \iint_R \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dA + i\iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dA.\end{aligned} \tag{3}$$

Because f is analytic in D, the real functions u and v satisfy the Cauchy-Riemann equations, $\partial u/\partial x = \partial v/\partial y$ and $\partial u/\partial y = -\partial v/\partial x$, at every point in D

Using the Cauchy-Riemann equations to replace $\partial u/\partial y$ and $\partial u/\partial x$ in (3) shows that

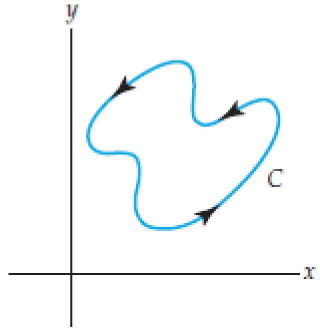
$$\begin{aligned}\oint_C f(z)\,dz &= \iint_R \left(-\frac{\partial v}{\partial x} + \frac{\partial v}{\partial x}\right) dA + i \iint_R \left(\frac{\partial v}{\partial y} - \frac{\partial v}{\partial y}\right) dA \\ &= \iint_R (0)\,dA + i \iint_R (0)\,dA = 0.\end{aligned}$$

Theorem Cauchy-Goursat Theorem

Suppose that a function f is analytic in a simply connected domain D. Then for every simple closed contour C in D, $\oint_C f(z)\,dz = 0$.

If f is analytic at all points within and on a simple closed contour C, then $\oint_C f(z)\,dz = 0$. (4)

EXAMPLE 1 Applying the Cauchy-Goursat Theorem



Evaluate $\oint_C e^z\,dz$, where the contour C is shown in Figure

Solution The function $f(z) = e^z$ is entire and consequently is analytic at all points within and on the simple closed contour C. It follows from the form of the Cauchy-Goursat theorem given in (4) that $\oint_C e^z\,dz = 0$.

The point of Example 1 is that $\oint_C e^z\,dz = 0$ for *any* simple closed contour in the complex plane. Indeed, it follows that for any simple closed contour C and any entire function f, such as $f(z) = \sin z$, $f(z) = \cos z$, and $p(z) = a_n z^n + a_{n-1} z^n + \cdots + a_1 z + a_0,\ n = 0, 1, 2, \ldots$,that

$$\oint_C \sin z\,dz = 0, \quad \oint_C \cos z\,dz = 0, \quad \oint_C p(z)\,dz = 0,$$

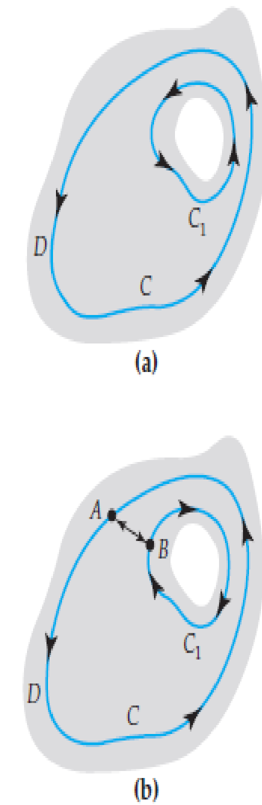
and so on.

EXAMPLE 2 Applying the Cauchy-Goursat Theorem

Evaluate $\oint_C \frac{dz}{z^2}$, where the contour C is the ellipse $(x-2)^2 + \frac{1}{4}(y-5)^2 = 1$.

Solution The rational function $f(z) = 1/z^2$ is analytic everywhere except at $z = 0$. But $z = 0$ is not a point interior to or on the simple closed elliptical contour C. Thus, from (4) we have that $\oint_C \frac{dz}{z^2} = 0$.

Cauchy-Goursat Theorem for Multiply Connected Domains If f is analytic in a multiply connected domain D then we cannot conclude that $\oint_C f(z)\,dz = 0$ for every simple closed contour C in D. To begin, suppose that D is a doubly connected domain and C and C_1 are simple closed contours such that C_1 surrounds the "hole" in the domain and is interior to C. See Figure (a). Suppose, also, that f is analytic on each contour and at each point interior to C but exterior to C_1. By introducing the crosscut AB shown in Figure (b), the region bounded between the curves is now simply connected. From (iv) of Theorem the integral from A to B has the opposite value of the integral from B to A, and so from (4) we have



Doubly connected domain D

$$\oint_C f(z)\,dz + \int_{AB} f(z)\,dz + \int_{-AB} f(z)\,dz + \oint_{C_1} f(z)\,dz = 0$$

or

$$\oint_C f(z)\,dz = \oint_{C_1} f(z)\,dz. \tag{5}$$

The last result is sometimes called the **principle of deformation of contours** since we can think of the contour C_1 as a continuous deformation of the contour C. Under this deformation of contours, the value of the integral does not change. In other words, (5) allows us to evaluate an integral over a complicated simple closed contour C by replacing C with a contour C_1 that is more convenient.

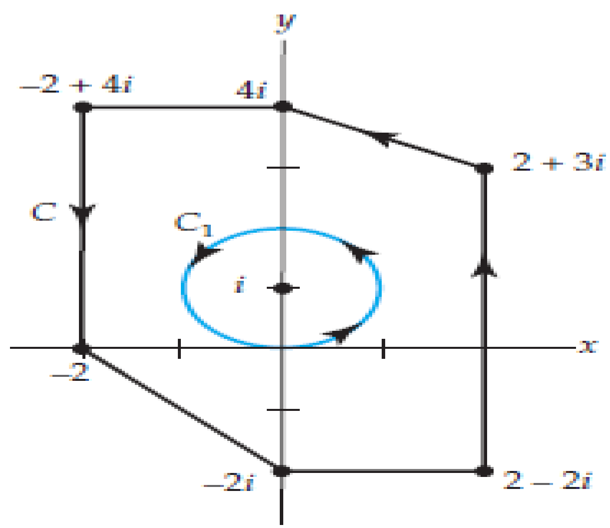
EXAMPLE 3 Applying Deformation of Contours

Evaluate $\oint_C \frac{dz}{z-i}$, where C is the contour shown in black in Figure

Solution In view of (5), we choose the more convenient circular contour C_1 drawn in color in the figure. By taking the radius of the circle to be $r = 1$, we are guaranteed that C_1 lies within C. In other words, C_1 is the circle

$|z - i| = 1$, which from (10) can be parametrized by $z = i + e^{it}$, $0 \le t \le 2\pi$. From $z - i = e^{it}$ and $dz = ie^{it}dt$ we obtain

$$\oint_C \frac{dz}{z-i} = \oint_{C_1} \frac{dz}{z-i} = \int_0^{2\pi} \frac{ie^{it}}{e^{it}}dt = i\int_0^{2\pi} dt = 2\pi i.$$



The result obtained in Example 3 can be generalized. Using the principle of deformation of contours (5) and proceeding as in the example, it can be shown that if z_0 is any constant complex number interior to *any* simple closed contour C, then for n an integer we have

$$\oint_C \frac{dz}{(z-z_0)^n} = \begin{cases} 2\pi i & n = 1 \\ 0, & n \neq 1. \end{cases} \tag{6}$$

The fact that the integral in (6) is zero when $n \neq 1$ follows only partially from the Cauchy-Goursat theorem. When n is zero or a negative integer, $1/(z-z_0)^n$ is a *polynomial* and therefore entire. Theorem and the discussion following Example 1 then indicates that $\oint_C dz/(z-z_0)^n = 0$. It is left as an exercise to show that the integral is still zero when n is a positive integer different from 1.

Analyticity of the function f at all points within and on a simple closed contour C is *sufficient* to guarantee that $\oint_C f(z)\,dz = 0$. However, the result in (6) emphasizes that analyticity is not *necessary*; in other words, it can happen that $\oint_C f(z)\,dz = 0$ without f being analytic within C. For instance, if C in Example 2 is the circle $|z| = 1$, then (6), with the identifications $n = 2$ and $z_0 = 0$, immediately gives $\displaystyle\oint_C \frac{dz}{z^2} = 0$. Note that $f(z) = 1/z^2$ is not analytic at $z = 0$ within C.

EXAMPLE 4 Applying Formula (6)

Evaluate $\oint_C \frac{5z+7}{z^2+2z-3}dz$, where C is circle $|z-2|=2$.

Solution Since the denominator factors as $z^2+2z-3=(z-1)(z+3)$ the integrand fails to be analytic at $z=1$ and $z=-3$. Of these two points, only $z=1$ lies within the contour C, which is a circle centered at $z=2$ of radius $r=2$. Now by partial fractions

$$\frac{5z+7}{z^2+2z-3}=\frac{3}{z-1}+\frac{2}{z+3}$$

and so
$$\oint_C \frac{5z+7}{z^2+2z-3}\,dz = 3\oint_C \frac{1}{z-1}\,dz + 2\oint_C \frac{1}{z+3}\,dz. \tag{7}$$

In view of the result given in (6), the first integral in (7) has the value $2\pi i$, whereas the value of the second integral is 0 by the Cauchy-Goursat theorem.

Hence, (7) becomes

$$\oint_C \frac{5z+7}{z^2+2z-3}\,dz = 3(2\pi i)+2(0)=6\pi i.$$

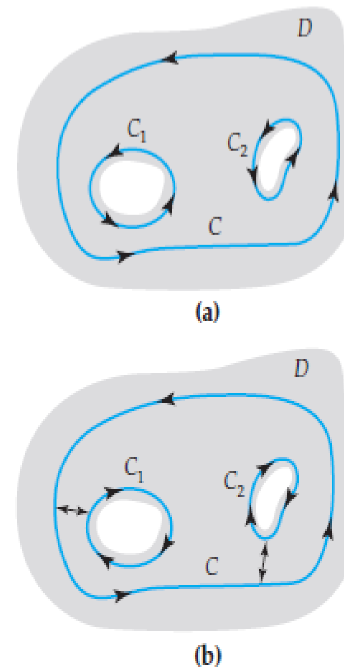
If C, C_1, and C_2 are simple closed contours as shown in Figure (a) and if f is analytic on each of the three contours as well as at each point interior to C but exterior to both C_1 and C_2, then by introducing crosscuts between C_1 and C and between C_2 and C, as illustrated in Figure (b), it follows from Theorem that

$$\oint_C f(z)\,dz + \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz = 0$$

and so

$$\oint_C f(z)\,dz = \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz. \tag{7}$$

The next theorem summarizes the general result for a multiply connected domain with n "holes."



Triply connected domain D

Theorem Cauchy-Goursat Theorem for Multiply Connnected Domains

Suppose C, $C_1, \ldots, C_n$ are simple closed curves with a positive orientation such that $C_1, C_2, \ldots, C_n$ are interior to C but the regions interior to each C_k, $k = 1, 2, \ldots, n$, have no points in common. If f is analytic on each contour and at each point interior to C but exterior to all the C_k, $k = 1, 2, \ldots, n$, then

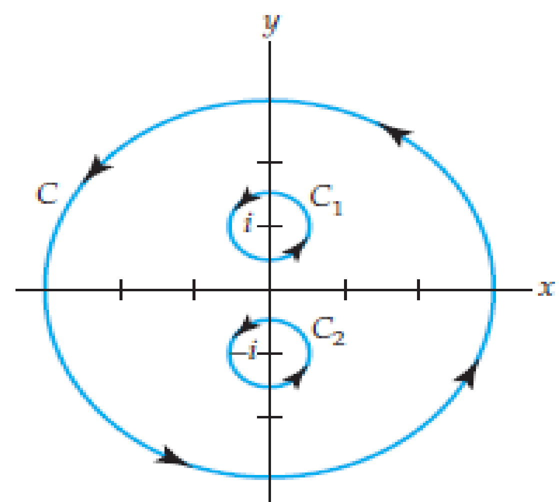
$$\oint_C f(z)\,dz = \sum_{k=1}^{n} \oint_{C_k} f(z)\,dz. \qquad (8)$$

EXAMPLE 5 Applying Theorem

Evaluate $\oint_C \frac{dz}{z^2+1}$, where C is the circle $|z| = 4$.

Solution In this case the denominator of the integrand factors as $z^2+1 = (z-i)(z+i)$. Consequently, the integrand $1/(z^2+1)$ is not analytic at $z = i$ and at $z = -i$. Both of these points lie within the contour C. Using partial fraction decomposition once more, we have

$$\frac{1}{z^2+1} = \frac{1}{2i}\frac{1}{z-i} - \frac{1}{2i}\frac{1}{z+i}$$



and
$$\oint_C \frac{dz}{z^2+1} = \frac{1}{2i}\oint_C \left[\frac{1}{z-i} - \frac{1}{z+i}\right] dz.$$

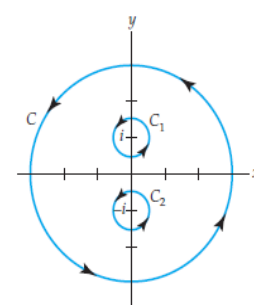
We now surround the points $z = i$ and $z = -i$ by circular contours C_1 and C_2, respectively, that lie entirely within C. Specifically, the choice $|z-i| = \frac{1}{2}$ for C_1 and $|z+i| = \frac{1}{2}$ for C_2 will suffice. From Theorem 5 we can write

$$\oint_C \frac{dz}{z^2+1} = \frac{1}{2i}\oint_{C_1} \left[\frac{1}{z-i} - \frac{1}{z+i}\right] dz + \frac{1}{2i}\oint_{C_2} \left[\frac{1}{z-i} - \frac{1}{z+i}\right] dz$$

$$= \frac{1}{2i}\oint_{C_1} \frac{1}{z-i} dz - \frac{1}{2i}\oint_{C_1} \frac{1}{z+i} dz + \frac{1}{2i}\oint_{C_2} \frac{1}{z-i} dz - \frac{1}{2i}\oint_{C_2} \frac{1}{z+i} dz. \quad (9)$$

Because $1/(z+i)$ is analytic on C_1 and at each point in its interior and because $1/(z-i)$ is analytic on C_2 and at each point in its interior, it follows from (4) that the second and third integrals in (9) are zero. Moreover, it follows from (6), with $n = 1$, that

$$\oint_{C_1} \frac{dz}{z-i} = 2\pi i \qquad \text{and} \qquad \oint_{C_2} \frac{dz}{z+i} = 2\pi i.$$



Thus (9) becomes

$$\oint_C \frac{dz}{z^2+1} = \pi - \pi = 0.$$

Let γ be the positively oriented circle with radius 1 and center i .

Evaluate the following counter integrals

a) $\oint_\gamma \bar{z}dz = ?$ **b)** $\oint_\gamma \frac{dz}{z^2-2} = ?$

$$a)\ \gamma : |z-i| = 1 \Rightarrow z - i = e^{it} \quad 0 \le t \le 2\pi$$

$$z = i + e^{it} \quad dz = ie^{it}dt \quad \bar{z} = -i + e^{-it}$$

$$\Rightarrow \oint_\gamma \bar{z}dz = \int_0^{2\pi} (-i + e^{-it})\, ie^{it}dt = \int_0^{2\pi} (e^{it} + i)dt$$

$$= \left| \frac{1}{i} e^{it} + it \right|_0^{2\pi} = 2\pi i$$

b) $\oint_\gamma \frac{dz}{z^2-2}$ $\quad f(z) = \frac{1}{z^2-2} = \frac{1}{(z-\sqrt{2})(z+\sqrt{2})}$.

f(z) is not analytic at $z_1 = \sqrt{2}$ ve $z_2 = -\sqrt{2}$

$$\left.\begin{aligned} |z_1 - i| &= |\sqrt{2} - i| = \sqrt{3} > 1 \\ |z_2 - i| &= |-\sqrt{2} - i| = \sqrt{3} > 1 \end{aligned}\right) \quad z_1 \text{ and } z_2 \text{ outside of } \gamma$$

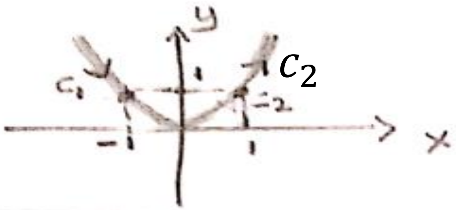
$$\Rightarrow \oint_\gamma \frac{dz}{z^2-2} = 0$$

a.) $f(x) = \begin{cases} 2 & x < 0 \\ 6x & x > 0 \end{cases}$ and

$C:$ from $z=-1+i$ to $z=1+i$ along the parabola $y=x^2$

$\int_C f(z)dz = ?$

<u>Solution:</u>



$y = x^2$, $dy = 2x\,dx$

$dz = dx + i\,dy$

$dz = dx + i2x\,dx = (1 + i2x)\,dx$

$$\int_C f(z)dz = \int_{C_1} f(z)dz + \int_{C_2} f(z)dz$$

$$\int_{C_1} f(z)dz = \int_{-1}^{0} 2\cdot(1+2ix)dx = (2x + 2ix^2)\Big|_{-1}^{0} = 2-2i$$

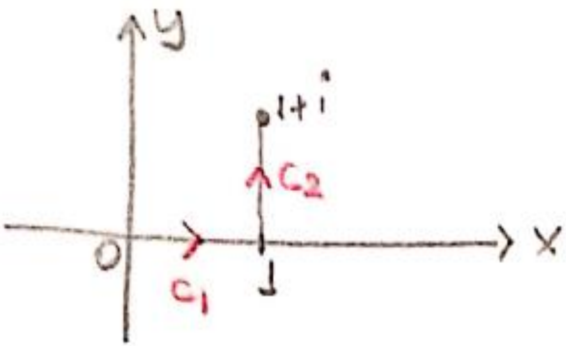
$$\int_{C_2} f(z)dz = \int_{0}^{1} 6x\cdot(1+2ix)dx = (3x^2 + 4ix^3)\Big|_{0}^{1} = 3+4i$$

$$\int_C f(z)dz = (2-2i) + (3+4i) = 5+2i$$

b) C: from $z=0$ to $z=1$ and from $z=1$ to $z=1+i$,

$$\int_C \sin(z)\,dz = ?$$

Solution:



$$\int_C \sin z\,dz = \int_{C_1} \sin z\,dz + \int_{C_2} \sin z\,dz$$

Let us consider C_1

C_1: $y=0 \Rightarrow dy=0$. Thus, $z = x + \cancel{iy}^{\to 0}$, that is $z=x$, so $dz=dx$

$$\int_{C_1} \sin z\,dz = \int_0^1 \sin(x+i0)\,dx = -\cos x \Big|_0^1 = -\cos 1 + 1$$

Let us consider C_2

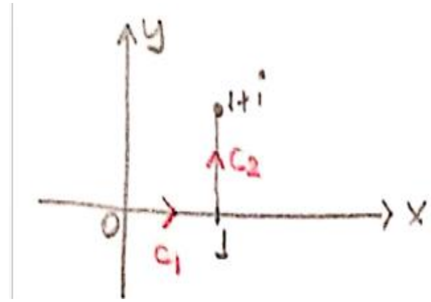
$C_2: \; x = 1 \Rightarrow dx = 0 \qquad \sin z = \sin(1+iy) \qquad z = 1+iy$

$dz = i\,dy$

$$\int_{C_2} \sin z \, dz = \int_0^1 \sin(1+iy).i\,dy = -\cos(1+iy)\Big|_0^1$$

$$= -\cos(1+i) + \cos 1$$

$$\int_C \sin z \, dz = -\cancel{\cos 1} + 1 - \cos(1+i) + \cancel{\cos 1}$$



$$= 1 - \cos(1+i)$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$\cos(1+i) = \frac{e^{i(1+i)} + e^{-i(1+i)}}{2} = \frac{1}{2}\left\{e^{-1}.e^{i} + e.e^{-i}\right\}$$

$$= \frac{1}{2}\left[e^{-1}(\cos 1 + i\sin 1) + e(\cos 1 - i\sin 1)\right]$$

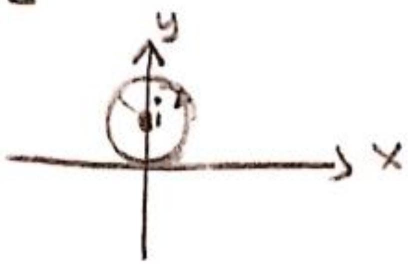
$$= \frac{1}{2}.\cos 1.(e^{-1} + e^{1}) + \frac{1}{2} i \sin 1 (e^{-1} - e^{1})$$

$$= \cos 1 . \text{ch} 1 - i \sin 1 . \text{sh} 1$$

$$\int_C \sin z \, dz = 1 - \cos 1 . \text{ch} 1 + i \sin 1 . \text{sh} 1$$

Evaluate the following integral

$$\int_C \bar{z}\, dz = ? \qquad \text{where} \quad C: |z-i|=1$$



$$z - i = 1.e^{i\theta}$$

$$z = i + e^{i\theta} \qquad \bar{z} = -i + e^{-i\theta}$$

$$dz = i e^{i\theta}\, d\theta$$

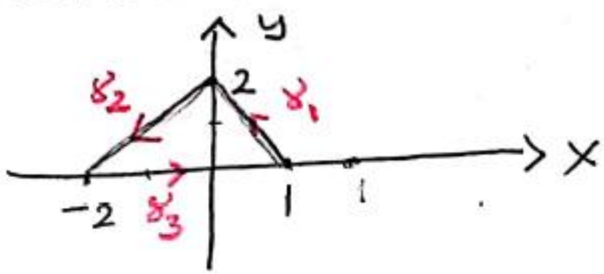
$$\int_C \bar{z}\, dz = \int_0^{2\pi} (-i + e^{-i\theta}) . i . e^{i\theta}\, d\theta$$

$$= \int_0^{2\pi} (e^{i\theta} + i)\, d\theta = \left(\frac{1}{i} e^{i\theta} + i\theta\right)\Bigg|_0^{2\pi}$$

$$= \frac{1}{i}\underbrace{(e^{2\pi i} - 1)}_{0} + 2\pi i = 2\pi i$$

$$e^{2\pi i} = 1$$

Using contour integration, evaluate $\int_\Gamma \bar{z}\,dz$ where Γ is the contour consisting of three straight line segments connecting the points $(1,0)$, $(0,2)$, $(-2,0)$ traversed in a counter-clockwise direction.



$\gamma_1 : 1 \rightarrow 2i$

$\gamma_2 : 2i \rightarrow -2$

$\gamma_3 : -2 \rightarrow 1$

$z_1(t) = (2i-1)t + 1 \quad , \; t : 0 \rightarrow 1$

$z_2(t) = (-2-2i)t + 2i \quad , \; t : 0 \rightarrow 1$

$z_3(t) = t \quad , \; t : -2 \rightarrow 1$

$$\int_{\gamma_1} \bar{z}\,dz = \int_0^1 \left[(-2i-1)t+1\right](2i-1)\,dt = \int_0^1 \left[(2i-1) + t\left((-1)^2-(2i)^2\right)\right] dt$$

$$= (2i-1)t\Big|_0^1 + 5\int_0^1 t\,dt = 2i-1+5\cdot\frac{1}{2} = 2i+\frac{3}{2}$$

$$\int_{\gamma_2} \bar{z}\,dz = \int_0^1 ((-2+2i)t-2i)\cdot(-2-2i)\,dt$$

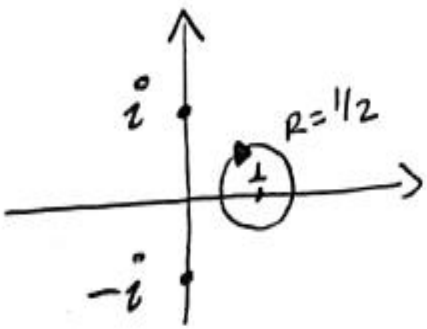
$$= \int_0^1 8i\,t\,dt + \int_0^1 (+4i-4)\,dt = 4i$$

$$\int_{\gamma_3} \bar{z}\,dz = \int_{-2}^1 t\,dt = \frac{t^2}{2}\Big|_{-2}^{1} = \frac{1}{2} - \frac{(-2)^2}{2} = -\frac{3}{2}$$

$$\int_{\Gamma} \bar{z}\,dz = \int_{\gamma_1} \bar{z}\,dz + \int_{\gamma_2} \bar{z}\,dz + \int_{\gamma_3} \bar{z}\,dz$$

$$= 2i + \cancel{\frac{3}{2}} + 4i - \cancel{\frac{3}{2}} = 6i$$

Evaluate $\int_C \frac{z+1}{z^2+1}\,dz$, where $C: 2|z-1|=1$ with negative orientation



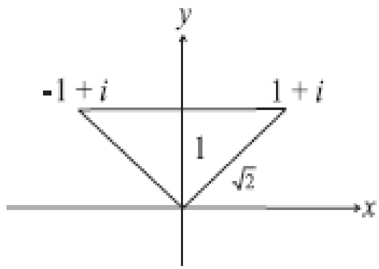
$$\int_C f(z)\,dz = \oint_C \frac{z+1}{(z-i)(z+i)}\,dz = 0$$

f is analytic in C

f is analytic on $\begin{pmatrix} \mathbb{C}-\{i\} \\ \mathbb{C}-\{-i\} \end{pmatrix}$

Show that

$\left| \int_C \frac{1}{z^2}\, dz \right| \leq 2$, where C is the line segment joining $-1+i$ and $1+i$.



Solution

Along the contour C, we have $z = x + i$, $-1 \leq x \leq 1$, so that $1 \leq |z| \leq \sqrt{2}$. Correspondingly, $\frac{1}{2} \leq \frac{1}{|z|^2} \leq 1$. Here, $M = \max_{z \in C} \frac{1}{|z|^2} = 1$ and the arc length $L = 2$. We have

$$\left| \int_C \frac{1}{z^2}\, dz \right| \leq ML = 2.$$

Estimate an upper bound of the modulus of the integral

$$I = \int_C \frac{\text{Log } z}{z - 4i}\, dz$$

where C is the circle $|z| = 3$.

Now, $\left|\dfrac{\text{Log } z}{z - 4i}\right| \leq \dfrac{\big|\ln|z|\big| + |\text{Arg } z|}{\big||z| - |4i|\big|}$ so that

$$\max_{z \in C} \left|\frac{\text{Log } z}{z - 4i}\right| \leq \frac{\ln 3 + \pi}{|3 - 4|} = \ln 3 + \pi; \quad L = (2\pi)(3) = 6\pi.$$

Hence, $\left|\displaystyle\int_C \frac{\text{Log } z}{z - 4i}\, dz\right| \leq 6\pi(\pi + \ln 3)$.

Find an upper bound for $\left|\int_\Gamma e^z/(z^2+1)\,dz\right|$, where Γ is the circle $|z|=2$ traversed once in the counterclockwise direction.

Solution

The path of integration has length $L=4\pi$. Next we seek an upper bound M for the function $e^z/(z^2+1)$ when $|z|=2$. Writing $z=x+iy$, we have

$$|e^z|=|e^{x+iy}|=e^x\le e^2, \quad \text{for} \quad |z|=\sqrt{x^2+y^2}=2,$$

and by the triangle inequality

$$|z^2+1|\ge|z|^2-1=4-1=3 \quad \text{for} \quad |z|=2.$$

Hence, $|e^z/(z^2+1)|\le e^2/3$ for $|z|=2$, and so

$$\left|\int_\Gamma \frac{e^z}{z^2+1}\,dz\right| \le \frac{e^2}{3}\cdot 4\pi.$$